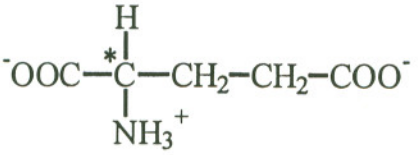
Biochemistry Cumulative Exam

Sept 18, 1999

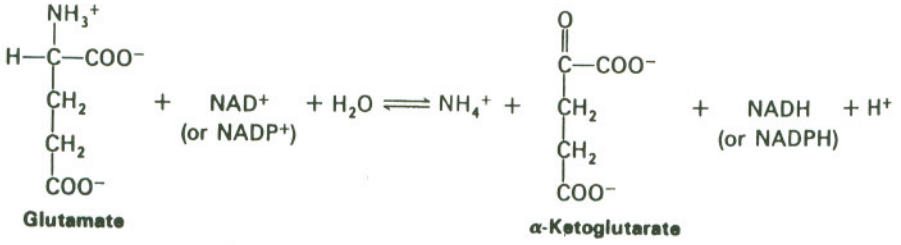
Glutamate radiolabeled with C-14 at carbon 2 is added to a liver cell suspension as the sole source of carbon and nitrogen. A short period later, glucose is isolated and analyzed to determine the <u>first</u> carbon atom(s) in the glucose which become labeled with C-14.

Glutamate: 

a) (40) Briefly outline the metabolic pathways needed for the conversion of glutamate to glucose under the above conditions.

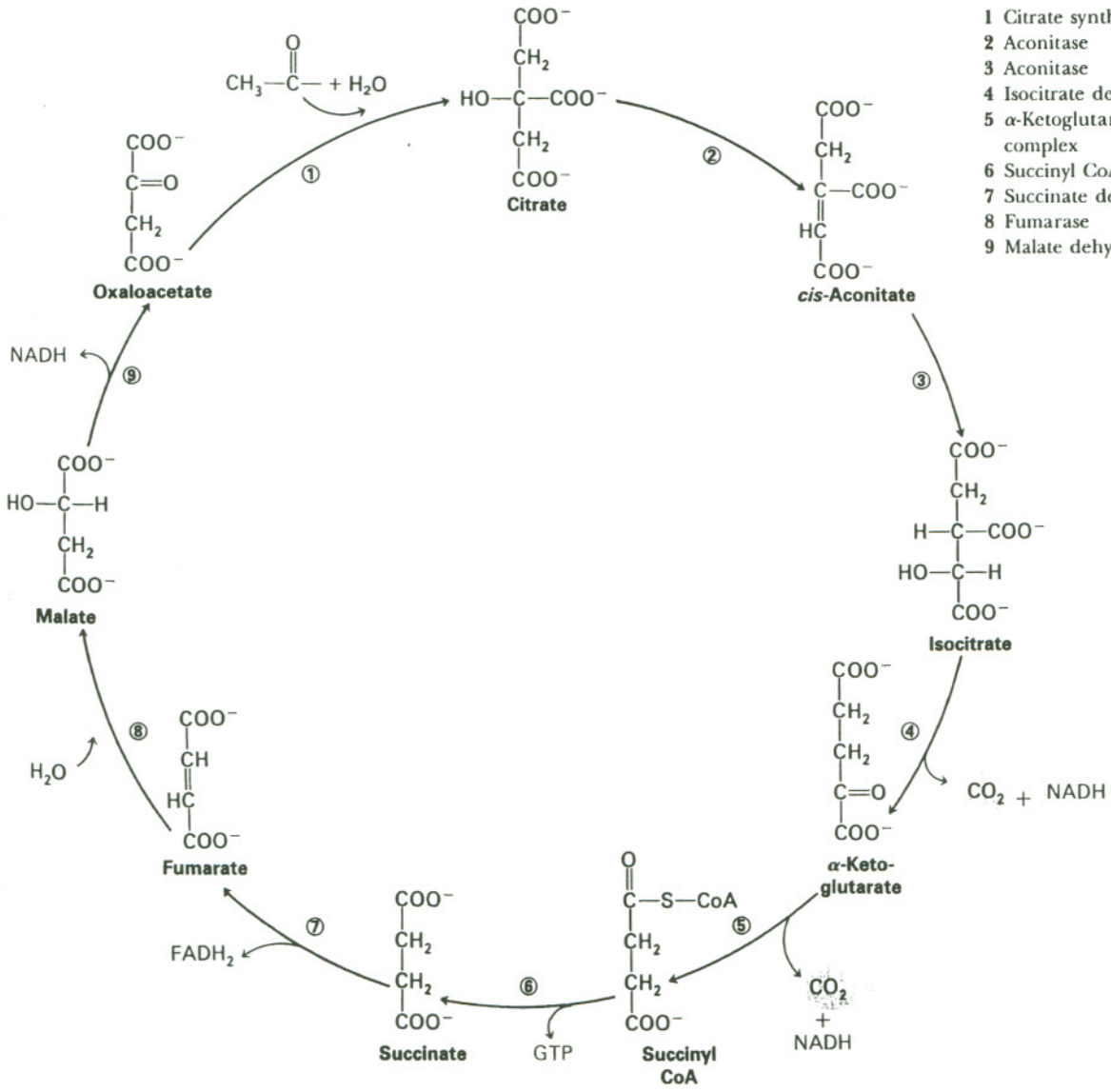
b) (30) Trace the C-14 atom on glutamate carbon number 2 through the above pathways and identify the <u>first</u> carbon atom(s) on glucose which will be labeled. Show the labeling of key intermediates along the pathway.

c) (30) Discuss the bioenergetics of the conversion of glutamate to glucose. Write a balanced equation for the conversion of glutamate to glucose and NH_4^+ in a liver cell under aerobic conditions where oxidative phosphorylation is functional. How many ATP molecules are required or generated for this reaction to occur in a liver cell under aerobic conditions?



following enzymes, as numbered in the diagram:

1 Citrate synthase
2 Aconitase
3 Aconitase
4 Isocitrate dehydrogenase
5 α-Ketoglutarate dehydrogenase complex
6 Succinyl CoA synthetase
7 Succinate dehydrogenase
8 Fumarase
9 Malate dehydrogenase



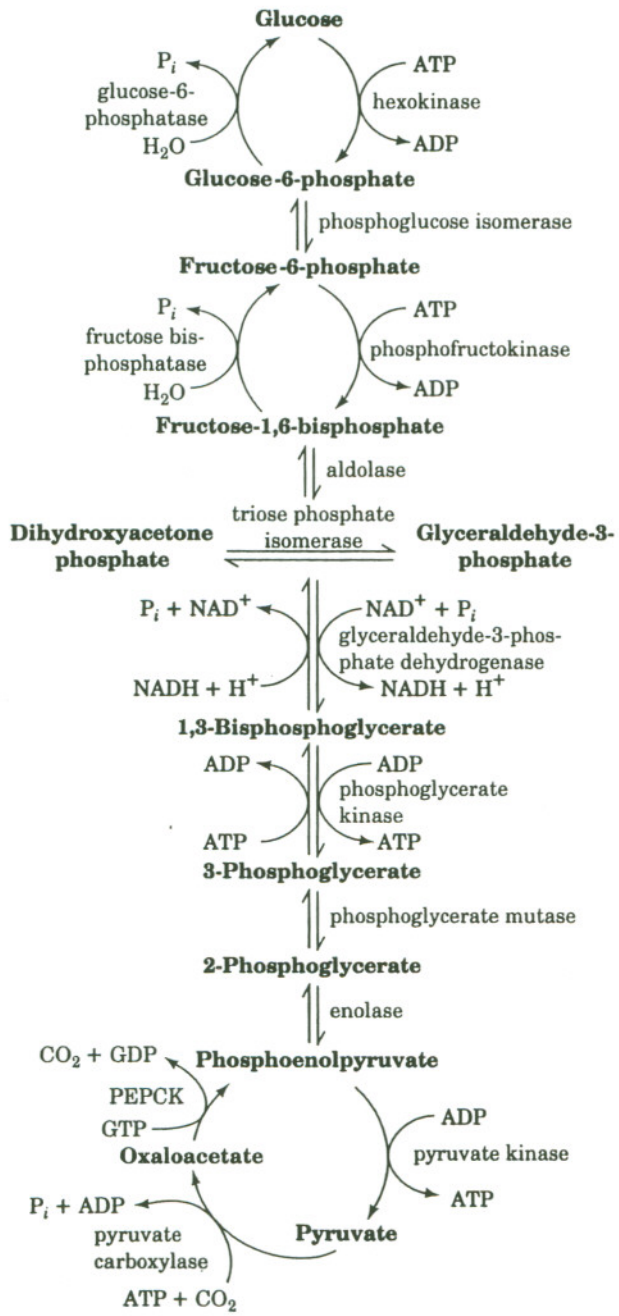
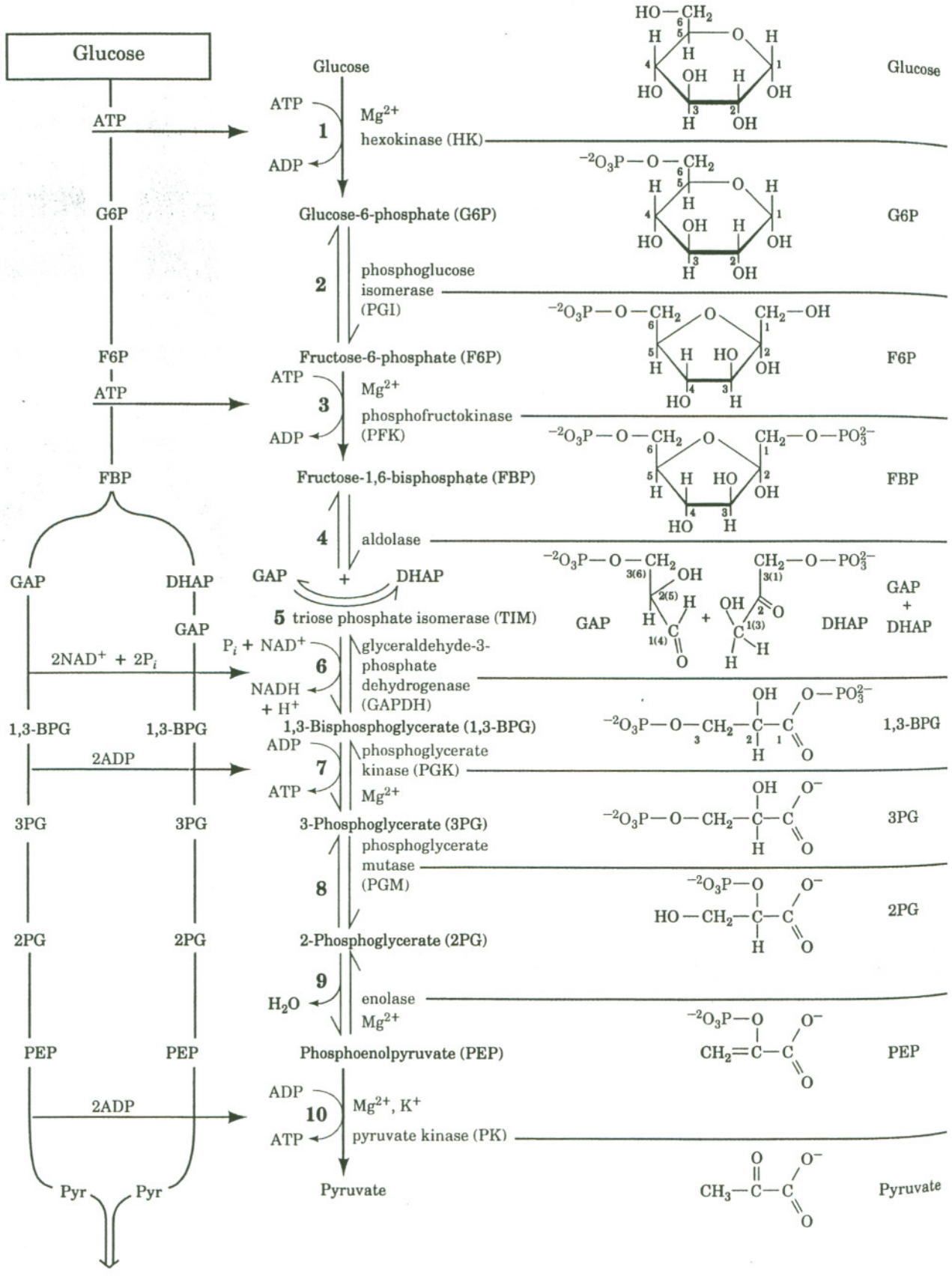


Figure 15-22. ***Key to Metabolism.*** **Comparison of the pathways of gluconeogenesis and glycolysis.** The red arrows represent the steps that are catalyzed by different enzymes in gluconeogenesis. The other seven reaction steps of gluconeogenesis are catalyzed by glycolytic enzymes that function near equilibrium.


Glucose
ATP
G6P
F6P
ATP
FBP
GAP
DHAP
GAP
2NAD+ + 2Pi
1,3-BPG
1,3-BPG
2ADP
3PG
3PG
2PG
2PG
PEP
PEP
2ADP
Pyr
Pyr
Glucose
ATP
1
Mg2+
hexokinase (HK)
ADP
Glucose-6-phosphate (G6P)
2
phosphoglucose isomerase (PGI)
Fructose-6-phosphate (F6P)
ATP
3
Mg2+
phosphofructokinase (PFK)
ADP
Fructose-1,6-bisphosphate (FBP)
4
aldolase
GAP + DHAP
5 triose phosphate isomerase (TIM)
Pi + NAD+
6
glyceraldehyde-3-phosphate dehydrogenase (GAPDH)
NADH + H+
1,3-Bisphosphoglycerate (1,3-BPG)
ADP
7
phosphoglycerate kinase (PGK)
ATP
Mg2+
3-Phosphoglycerate (3PG)
8
phosphoglycerate mutase (PGM)
2-Phosphoglycerate (2PG)
9
H2O
enolase
Mg2+
Phosphoenolpyruvate (PEP)
ADP
10
Mg2+, K+
pyruvate kinase (PK)
ATP
Pyruvate
Glucose
G6P
F6P
FBP
GAP
DHAP
GAP + DHAP
1,3-BPG
3PG
2PG
PEP
Pyruvate